

APPENDIX

A. Formula and Definition

1. Euler Expansion: $e^{j\pm x} = \cos x \pm j \sin x$.
2. Continuous-time Convolution: $x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$
3. Discrete-time Convolution: $x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$
4. Continuous-time Fourier Series: $x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$, $a_k = \frac{1}{T} \int_T x(t)e^{-jk(2\pi/T)t} dt$
5. Discrete-time Fourier Series: $x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$, $a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n]e^{-jk(2\pi/N)n}$
6. Continuous-time Fourier Transform: $x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t} d\omega$, $X(j\omega) = \int_{-\infty}^{+\infty} x(t)e^{-j\omega t} dt$
7. CT Periodic Signal Fourier Transform: $x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$, $X(j\omega) = 2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$
8. CT Aperiodic Signal Parseval Theorem: $E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega < \infty$

B. Convolution Result

1. $(u(t+T) - u(t-T)) * (u(t+T) - u(t-T)) = (2T - |t|)(u(t+2T) - u(t-2T))$ (Triangular wave!)
2. $e^{-at}u(t) * u(t) = \frac{1-e^{-at}}{a}u(t)$

C. Fourier Transform

1. $\delta(t-t_0) \xrightarrow{\mathcal{F}} e^{-j\omega t_0}$ $u(t) \xrightarrow{\mathcal{F}} \frac{1}{j\omega} + \pi\delta(\omega)$ $e^{j\omega_0 t} \xrightarrow{\mathcal{F}} 2\pi\delta(\omega - \omega_0)$
2. $\cos \omega_0 t \xrightarrow{\mathcal{F}} \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$ $\sin \omega_0 t \xrightarrow{\mathcal{F}} \frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
3. $x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases} \xrightarrow{\mathcal{F}} \frac{2 \sin \omega T_1}{\omega}$ $\frac{\sin Wt}{\pi t} \xrightarrow{\mathcal{F}} X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases}$
4. $e^{-at}u(t), \text{Re}\{a\} > 0 \xrightarrow{\mathcal{F}} \frac{1}{a+j\omega}$ $te^{-at}u(t), \text{Re}\{a\} > 0 \xrightarrow{\mathcal{F}} \frac{1}{(a+j\omega)^2}$
5. $\frac{t^{n-1}}{(n-1)!}e^{-at}u(t), \text{Re}\{a\} > 0 \xrightarrow{\mathcal{F}} \frac{1}{(a+j\omega)^n}$
6. $\sum_{n=-\infty}^{+\infty} \delta(t-nT) \xrightarrow{\mathcal{F}} \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - \frac{2\pi k}{T})$

D. Properties of Fourier Transform

1. CT Time Shifting and Frequency Shifting: $x(t-t_0) \xrightarrow{\mathcal{F}} e^{-j\omega t_0} X(j\omega)$ $e^{j\omega_0 t} x(t) \xrightarrow{\mathcal{F}} X(j(\omega - \omega_0))$
2. CT Time and Frequency Scaling: $x(at) \xrightarrow{\mathcal{F}} \frac{1}{|a|} X(\frac{j\omega}{a})$
3. CT Differentiation and Integration: $\frac{d}{dt}x(t) \xrightarrow{\mathcal{F}} j\omega X(j\omega)$ $\int_{-\infty}^t x(\tau)d\tau \xrightarrow{\mathcal{F}} \frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
4. Differentiation in Frequency : $tx(t) \xrightarrow{\mathcal{F}} j \frac{dX(j\omega)}{d\omega}$ $nx[n] \xrightarrow{\mathcal{F}} j \frac{dX(e^{j\omega})}{d\omega}$